EE 121B Fall 2026
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Homework Assignment 1 Solution

1. Energy band model

a) For silicon, if E_G decreases by 0.078 eV, by what fraction does n_i increase (assume T is constant at 300K)?

$$n_i = \sqrt{N_c N_v} e^{-E_G/2kT}$$

T is constant, N_c and N_v are constant.

$$\frac{n_{i2}}{n_{i1}} = \frac{e^{\frac{E_{G2}}{2kT}}}{e^{\frac{E_{G1}}{2kT}}} = e^{\frac{E_{G1}-E_{G2}}{2kT}} = e^{\frac{0.078}{2\times 0.026}} = 4.48$$

So n_i increase by 348%

b) If the temperature rises from 300K to 600K, by what additional fraction does n_i increase (assume E_G is constant = 1.068 eV, you do not need to worry about the decrease from problem a)?

$$N_c = 2\left(\frac{2\pi m^*_{n,DOS} kT}{h^2}\right)^{3/2} \quad N_v = 2\left(\frac{2\pi m^*_{p,DOS} kT}{h^2}\right)^{3/2},$$

$$\frac{n_{i2}}{n_{i1}} = \frac{\sqrt{N_{c2}N_{v2}} e^{\frac{-E_G}{2kT_2}}}{\sqrt{N_{c1}N_{v1}} e^{\frac{-E_G}{2kT_1}}}$$

$$= \left(\frac{600}{300}\right)^{\frac{3}{2}} e^{\frac{-1.068}{2k600} - \frac{-1.068}{2k300}}$$

$$= 84846$$

So n_i increase by 8500000%

c) If we further increase the temperature, Si starts exhibiting semi-metal properties. Think of a physical reason for this phenomenon. (Assume you haven't melted it).

When a semiconductor is heated, the lattice expands and the atomic distance increases. This causes the bandgap to shrink. The semiconductor will begin to behave more like a semi-metal.

2. Fermi-Dirac distribution

a) Under equilibrium condition, what is the probability of an electron state being occupied if it is located at the Fermi level?

Assume $E = E_f$, $f(E) = 0.5$.

b) If E_F is positioned at E_C, determine the probability of finding electrons in states at $E_C + kT$. (A numerical answer is required.)

$E_f = E_c$ and $E = E_c + kT$

$$f(E) = \frac{1}{1+e^{[(E_C+kT)-E_C]/kT}} = \frac{1}{1+e} = 0.27$$

c) The probability of a state being filled at E_C + kT is equal to the probability of a state being empty at E_C + 3 kT. Where is the Fermi level located?

$f(E_c + kT) = 1 - f(E_c + 3kT)$

$$\frac{1}{1+e^{\left[(E_C+\mathrm{kT})-E_f\right]/kT}} = 1-\frac{1}{1+e^{\left[(E_C+3\mathrm{kT})-E_f\right]/kT}}$$

$$\text{RHS} = \frac{1+e^{\left[(E_C+3\mathrm{kT})-E_f\right]/kT}-1}{1+e^{\left[(E_C+3\mathrm{kT})-E_f\right]/kT}} = \frac{e^{\left[(E_C+3\mathrm{kT})-E_f\right]/kT}}{1+e^{\left[(E_C+3\mathrm{kT})-E_f\right]/kT}} = \frac{1}{1+e^{-\left[(E_C+3\mathrm{kT})-E_f\right]/kT}}$$

$$\text{LHS} = \frac{1}{1+e^{\left[(E_C+\mathrm{kT})-E_f\right]/kT}}$$

$$\text{RHS = LHS: } \frac{1}{1+e^{\left[(E_C+\mathrm{kT})-E_f\right]/kT}} = \frac{1}{1+e^{-\left[(E_C+3\mathrm{kT})-E_f\right]/kT}}$$

$$(E_c + \mathrm{kT}) - E_f = -[(E_c + 3\mathrm{kT}) - E_f]$$

$$E_f = E_c + 2kT$$

3. Carrier Concentrations

Consider a Si sample under equilibrium conditions, doped with Boron to a concentration 10^{17} cm^{-3}.

a) At T = 300K, is this material n-type or p-type? What are the majority and minority carrier concentrations?

b) As the temperature of this sample is increased, n_i will eventually increase to be higher than the dopant concentration, and the sample will become intrinsic ($n \cong p \cong n_i$). Estimate the temperature at which this occurs, by finding the temperature at which n_i be much greater (at least 10× higher) than the dopant concentration. (Use the plot of n_i vs. T)

a) Boron (B) is a Group III element with three valence electrons, which is one less than Si. When a B atom replaces a Si atom, an electron from a nearby covalent bond can easily move to the B atom to fill its outmost electron shell, creating a hole. (The B atom becomes negatively ionized, as a result.) Assuming that each B atom results in one hole, the hole concentration (p) will be 10^{17} cm^{-3}. Since the np product is equal to n_i^2 under equilibrium conditions, the electron concentration will be $n_i^2/p = 10^{20}/10^{17} = 10^3$ cm^{-3}. Since $p > n$, this material is **p-type**. **The majority carrier (hole) concentration is 10^{17} cm^{-3} and the minority carrier (electron) concentration is 10^3 cm^{-3}.**

b) From the n_i *vs.* T plot, $n_i = 10^{18}$ cm^{-3} at **T = 1000K (727^0C)**.

4. Mobility and Drift

a) For some resistor applications, it is important to maintain a fixed value of resistance over a range of operating temperatures. Explain briefly how you could minimize variations (resulting from changes in temperature) in the resistivity of Si at temperatures near 300K. Assume that you are free to choose the value of resistivity.

b) A (non-compensated) p-type silicon sample is maintained at room temperature. When an electric field with a strength of 1000 V/cm is applied to the sample, the hole drift velocity is 1.75×10^5 cm/sec.

i) What is the mean free path of a hole in this sample? (Note: 1 kg cm^2/V s/C = 10^{-4} sec)

ii) Estimate the electron and hole concentrations in this sample.

a) Resistivity is dependent on carrier mobility and carrier concentration. Assuming that carrier concentration is relatively insensitive to variations in temperature around 300K, the temperature dependence of resistivity will be dictated by the temperature dependence of carrier mobility. Therefore, in order to minimize variations in the resistivity (cf. Lecture 2 Slide 18) with changes in temperature around 300K, we need to minimize variations in carrier mobility with changes in temperature around 300K. From Lecture 4 Slide 17 it can be seen that electron mobility is relatively insensitive to variation in temperature around 300K, if the dopant concentration is in the range 10^{18}-10^{19} cm^{-3}. Thus, **doping Si n-type to a concentration close to 10^{19} cm^{-3} will minimize variations in resistivity resulting from changes in temperature around 300K.**

b) Since hole drift velocity $v_{dp} = \mu_p \cdot \mathcal{E}$, $\mu_p = v_{dp} / \mathcal{E}$ $= 1.75\times10^5/1000 = 175$ cm^2/Vs.

i) Since $\mu_p = q\tau/m_p^*$, where τ is the average time between scattering events and m_p^* is the hole conductivity effective mass, the mean free path l can be calculated as

$$l = \tau v_{th} = (\mu_p m_p^*/q) \cdot v_{th} = 175\times0.39\times9.1\times10^{-31}\times1.9\times10^7/(1.6\times10^{-19}) \times10^{-4} \text{ cm} = 0.007\times10^{-4} \text{ cm} = \mathbf{7 \text{ nm}}$$

ii) From Lecture 4 Slide 16 the dopant concentration (N_A) is ~10^{17} cm^{-3} for hole mobility = 350 cm^2/Vs. Thus, $\boldsymbol{p = 10^{18}\ cm^{-3}}$ and $\boldsymbol{n = n_i^2/p = 10^2\ cm^{-3}}$.

5. Resistivity

a) Consider a Si sample maintained at $T = 300$K under equilibrium conditions, doped with Boron to a concentration 2×10^{16} cm^{-3}:

i) What are the electron and hole concentrations (n and p) in this sample? Is it n-type or p-type?

ii) Suppose the sample is doped additionally with Phosphorus to a concentration 6×10^{16} cm^{-3}. Is the material now n-type or p-type? What is the resistivity of this sample?

b) Ultra-thin semiconductor materials are of interest for future nanometer-scale transistors, but can present undesirably high resistance to current flow. How low must the resistivity of a semiconductor material be, to ensure that the resistance of a 2nm-thick, 10nm-long, 100nm-wide region does not exceed 100 ohms? If this material is n-type Si, estimate the dopant concentration that would be needed to achieve this resistivity.

a) i) Boron is an acceptor in Si, so the material is **p-type** with $\boldsymbol{p = N_A = 2\times10^{16}\ cm^{-3}}$ and electron concentration $\boldsymbol{n = n_i^2/p = 5\times10^3\ cm^{-3}}$.

ii) Phosphorus is a donor in Si. Since $N_D > N_A$, the material is **n-type**.

The carrier concentrations are determined by the net dopant concentration, $N_D - N_A = 4\times10^{16}$ cm^{-3}, so that $\boldsymbol{n = 4\times10^{16}\ cm^{-3}}$ and $\boldsymbol{p = n_i^2/n = 2500\ cm^{-3}}$.

The carrier mobilities are determined by the total dopant concentration, $N_D + N_A = 8\times10^{16}$ cm^{-3}. From the plot on Slide 35 of Lecture 3, electron mobility $\mu_n \cong 820$ cm^2/Vs and hole mobility $\mu_p \cong 370$ cm^2/Vs.

Since $n >> p$, resistivity $\boldsymbol{\rho \cong} 1/qn\mu_n = 1/[(1.6\times10^{-19})(4\times10^{16})(820)] = \mathbf{0.19\ \Omega\text{-cm}}$

b) In order for the resistance ($R=\rho L/Wt$) of a semiconductor region to be no larger than 100 ohms, its **resistivity must not exceed** $100 \cdot Wt/L = 100\Omega \times 100\text{nm} \times 2\text{nm}/10\text{nm} = 2000\ \Omega\text{-nm} = \mathbf{2 \times 10^{-4}\ \Omega\text{-cm}}$.

From Slide 39 of Lecture 3, the dopant concentration that would be needed for n-type Si to have such a low resistivity is **$\sim 6 \times 10^{20}\ \text{cm}^{-3}$**.